

AI IN INDUSTRY · ADNI PROJECT

Anomaly, Risk, and Decision Support for Alzheimer's Care

A multimodal pipeline on the ADNI dataset: anomaly detection, conversion risk, and fairness-aware decision support

PART I: FOUNDATIONS

Data exploration, and data treatment

What ADNI actually contains, what it doesn't, and the decisions made before any model was trained.

What is ADNI

- ADNI, the Alzheimer's Disease Neuroimaging Initiative, is a long running, multi site study that has followed thousands of participants since October 2004 to find biomarkers that catch Alzheimer's disease early and track how it progresses.
- It is a public private partnership, funded by the National Institute on Aging together with pharmaceutical companies and foundations, run across more than 60 clinical sites in the United States and Canada.
- It has grown across four phases, ADNI 1 through ADNI 4, each one adding participants and refining what gets measured.

How ADNI is structured and used

- Every participant is placed in one of three groups, cognitively normal, mild cognitive impairment, or Alzheimer's dementia, then followed over years of repeat visits to see whether and when their diagnosis changes.
- Each visit can collect brain imaging (PET and MRI), blood and spinal fluid biomarkers, genetic data, and standard cognitive tests, the same modalities behind the six raw tables this project starts from.
- Because ADNI tracks the same people over time rather than comparing separate groups once, it is what makes a longitudinal question like conversion risk possible to ask at all.

The project in one paragraph

1

Three stages

Anomaly detection (Point 1), conversion-risk survival model (Point 2), decision support (Point 3)

2

Point 3 makes the risk score actionable

A recommended check-in interval, a fairness audit of that recommendation, and an explanation of which biomarkers drive it.

3

A pipeline extension chains all three

This session's focus: wiring the three stages into one genuine sequential chain, not three outputs sharing a table by coincidence.

Six raw tables (out of ADNI) on visits that anchor the project

**Diagnostic
Summary
15,931**

DXSUM rows
3,788 unique patients
(diagnostic backbone)

**Amyloid PET
Analysis
4,582**

UCBERKELEY_AMY
_6MM rows
amyloid PET, 2,177
patients

**MRI Analysis
12,151**

UCSFFSX7 rows
structural MRI, 3,231
patients

**CSF Biomarkers
3,174**

UPENNBBIOMK_ROC
HE rows
CSF markers, 1,660
patients

Plus PTDEMOG (demographics, 6,222 rows) and DATADIC.

Only 1,333 of 3,788 patients (35%) have all five modalities at once, a hard ceiling on any complete-case model.

Data quality issues found and fixed

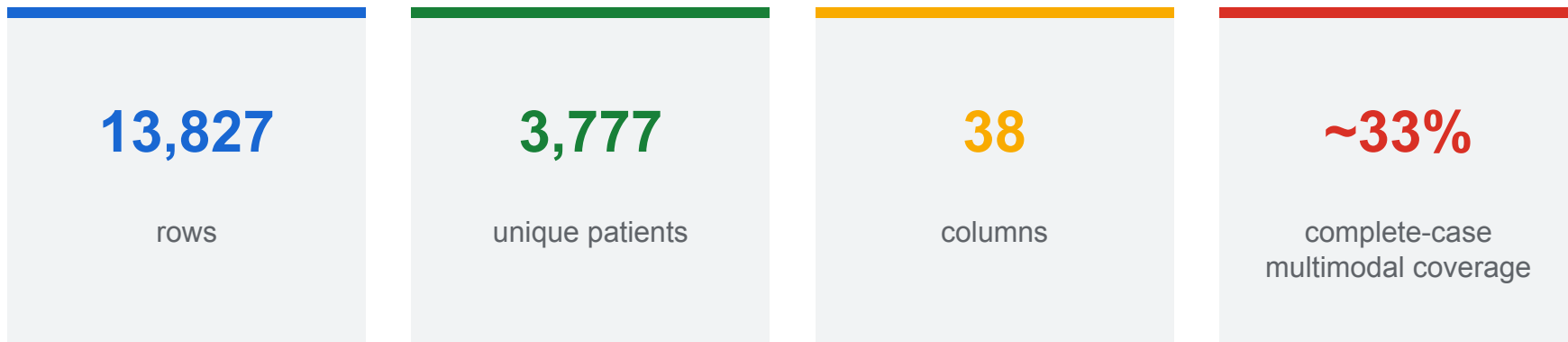
Coding & key issues

- ADNI sentinel codes (-1/-4) in demographics must be recoded to NaN before any statistic
- VISCODE vs VISCODE2 disagree in 62.3% of DXSUM rows, VISCODE2_norm used as the canonical key everywhere
- HAS_QC_ERROR tested as a blanket filter, populated for only 2,330/15,931 rows, always 0. Ruled out

Modality-specific issues

- Amyloid qc_flag: ~13% of PET scans fail QC (597 fail + 52 borderline of 4,582)
- MRI OVERALLQC missing for 91% of rows. STATUS=='complete' used instead. 737 rows have both 3T and 1.5T for one visit, 3T preferred.
- CSF: A β 42 ~99% populated but A β 40 only 29% so A β 42/40 ratio only usable for a minority.

The join: final.csv



final.csv merges the five modalities on RID + VISCODE2_norm. Patient count (3,777) closely matches the raw-table count (3,788), confirming the join isn't silently losing patients.

What the joined data actually supports

Visit depth & conversion

- Median 3 visits/patient, mean 3.66, max 20
- 1,474 patients (39%) appear only once, no trajectory available
- Of 2,303 patients with ≥ 2 diagnosed visits, only 590 (25.6%) ever worsen, 74.4% censored
- CN converts at 21.2% while MCI at 38.5%

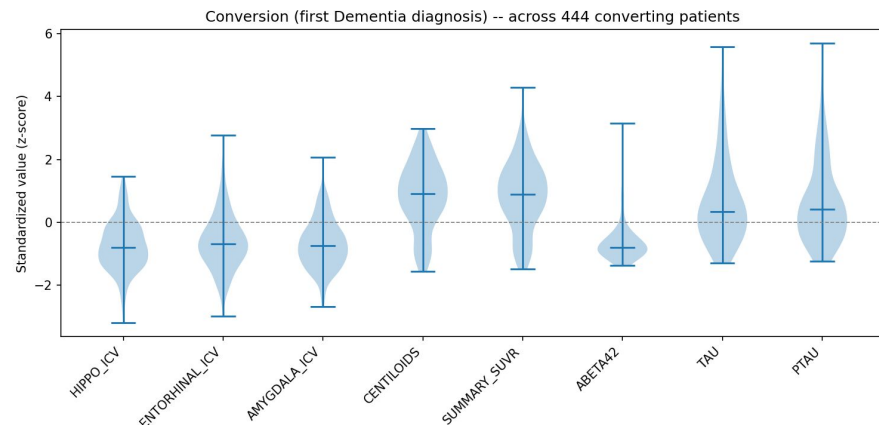
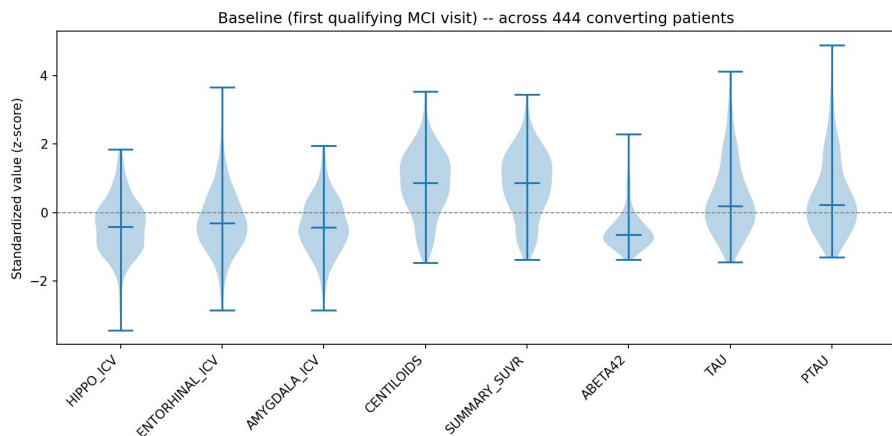
Coverage & preclinical signal

- Amyloid PET coverage is diagnosis-dependent: ~42% CN, ~29% MCI, ~22% Dementia
- MRI coverage is more uniform: 73–77% across all diagnoses
- ~1 in 3 CN patients with known amyloid status are amyloid-positive (701/2,200), biomarker-positive before symptoms
- This preclinical population is exactly what Point 1's anomaly detection targets

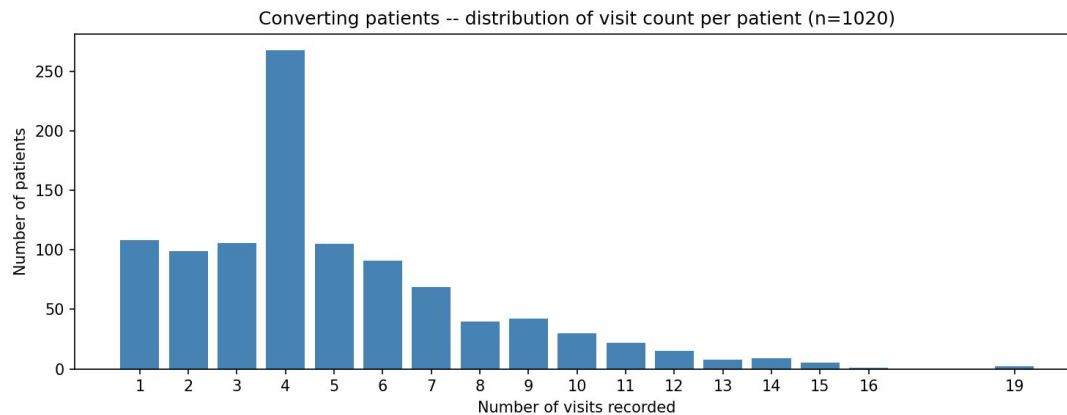
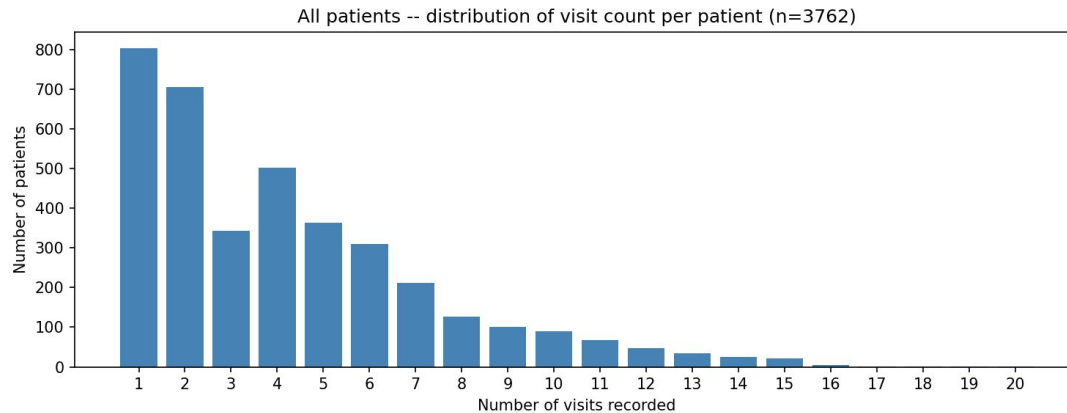
Shared schema for the pipeline

- One table keyed by RID + VISCODE2_norm, built from final.csv
- Point 1 contributes ANOMALY_SCORE and HI_SLOPE
- Point 2 contributes HAZARD_ESTIMATE and multi-horizon forecast columns (3/6/12/24 months)
- Point 3 contributes RECOMMENDED_INTERVAL and attribution columns
- A single RID-level train/validation/test split used consistently across every stage, so a leak in one stage cannot silently flow into the next

Distribution of main features across phases



Length of visits records



PART II: POINT 1

Anomaly Detection

Does a patient's visit look anomalous relative to a healthy reference population — without ever training on a diagnosis label?

Why frame this as anomaly detection?

- Goal: flag patients who look abnormal relative to cognitively-normal (CN) peers, without supervision from a diagnosis label.
- This targets the preclinical-AD population found in the data exploration: patients who are biomarker-positive but not yet symptomatic.
- A purely unsupervised approach is deliberately chosen. The CN is the only label used, as a reference population, not a training target.
- Question asked of every method: how well does the resulting anomaly score separate CN from MCI, and CN from AD, on held-out patients?

Three independent density models, one longitudinal comparison



PCA / Hotelling's T^2 + Q

The score is obtained as combined in-subspace (T^2) + out-of-subspace (Q/SPE) deviation.



Gaussian Mixture Model (GMM)

The score is the negative log-likelihood under the fitted CN mixture.



Autoencoder health index

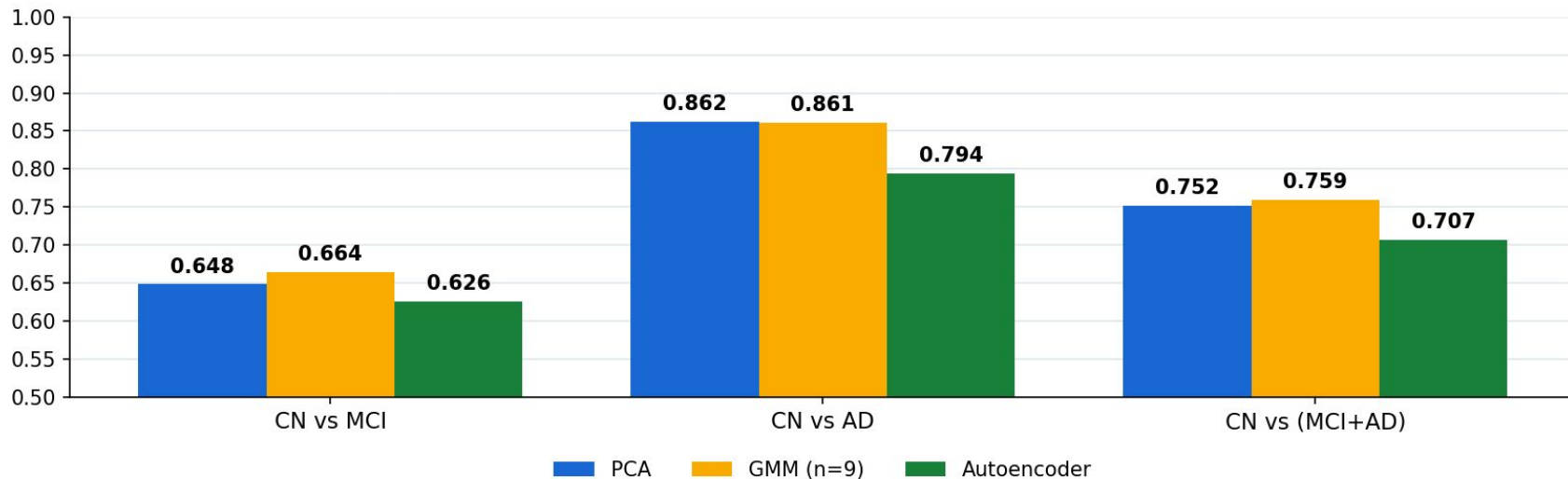
The score is the reconstruction error (MSE between original and reconstructed image).

Configurations and parameters, in full

Method	Model selection	Anomaly score
PCA / Hotelling's T^2+Q	First number of components clearing a 90% variance target.	$T^2/T^2_UCL + Q/Q_UCL$ ($T^2_UCL=29.32$, $Q_UCL=5.95$)
GMM	BIC based grid search: number of components (1 to 10) × covariance matrix either diag or full → 9 components, full covariance (BIC=103027.9)	−log-likelihood under the fitted CN mixture.
Autoencoder	Training stops automatically once improvement on a held-out 15% slice stalls for 30 straight epochs (up to 500 allowed). It stopped at epoch 324, keeping the best-performing weights.	Per-visit mean squared reconstruction error.

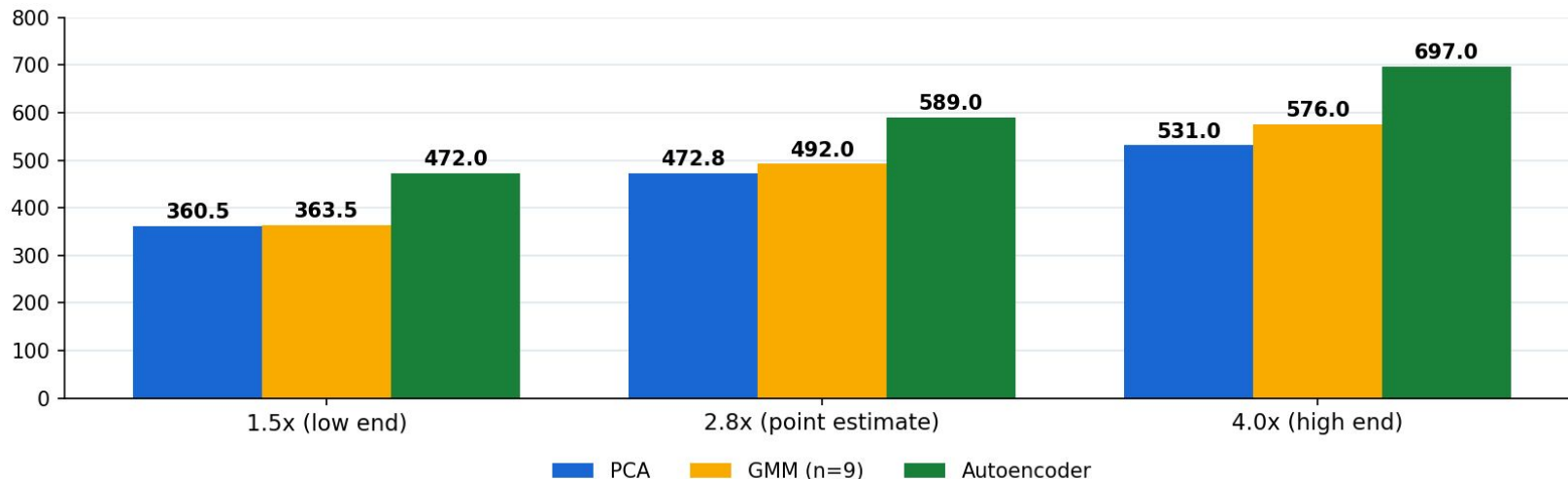
All three trained on the same CN-only visits (823 patients, 3,471 visits, 22 features). Model selection never touches an MCI/AD label.

One score per visit, evaluated at each patient's last visit



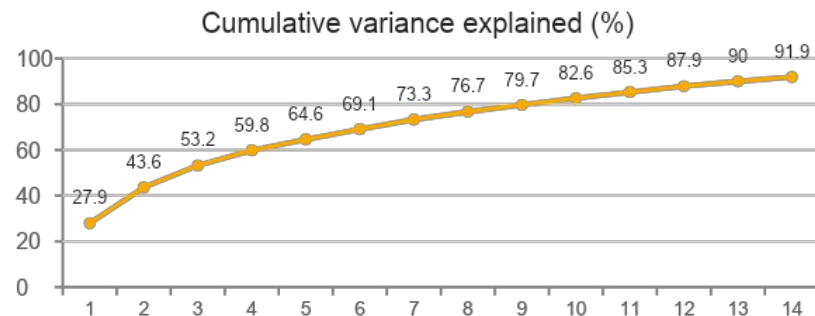
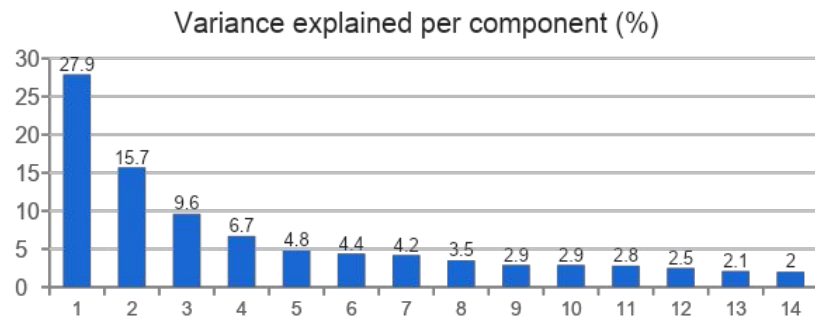
2,093 patients, 9,922 visits (823 CN patients / 3,471 CN visits fit each method). The corrected, per-visit cohort this project standardizes on. AUC uses each patient's last visit only. PCA leads on all three comparisons, most clearly CN vs AD (0.862 vs 0.847 vs 0.794).

From a score to a decision: choosing the threshold



CN vs AD, last visit only (679 CN / 685 AD), total cost at each method's own optimal threshold. Threshold picked by line search over percentile cutoffs of the CN training distribution, minimizing total cost without an arbitrary percentile.

What are PCA's 14 directions, roughly?



PC1 — 27.9%

Amygdala and hippocampus volume moving with temporal/fusiform cortical thickness which is a broad medial-temporal atrophy axis. Far out here = smaller, thinner memory-critical structures overall.

PC2 — 15.7%

Overall brain size (ICV) and entorhinal volume moving with ventricle size, correlated with sex. Mostly a body/head-size axis, not itself a disease signal.

PC3 — 9.6%

Enlarged ventricles moving with precuneus and temporal cortical thickness. An atrophy pattern distinct from PC1, centered on a different, earlier-affected region (precuneus).

PC4 — 6.7%

Entorhinal thickness/volume traded off against hippocampus and amygdala volume. The earliest AD-affected region weighed against the classic medial-temporal pair.

PC1–PC4 alone (~60% of variance) already read like a checklist of AD-relevant regions, amygdala, hippocampus, entorhinal cortex, ventricles, precuneus, without ever being told which regions matter.

Why PCA is the method carried forward

- AUC: GMM beats PCA on CN vs. MCI and in CN vs. (MCI+AD) while being approximately even on CN vs. AD.
- Cost: PCA also has the lowest total cost at every literature-anchored ratio (e.g. 531.0 vs. GMM's 576.0 at the 4.0× high end).
- Explainability: PCA's T^2 and Q scores decompose exactly into per-feature contributions that sum back to the patient's own score. The GMM has no comparably simple decomposition, and the Autoencoder would need an approximate, model-agnostic method like SHAP instead of an exact one.
- A clean result on all three criteria, not a trade-off between them. PCA is ahead on accuracy, cost, and explainability.

POINT 2

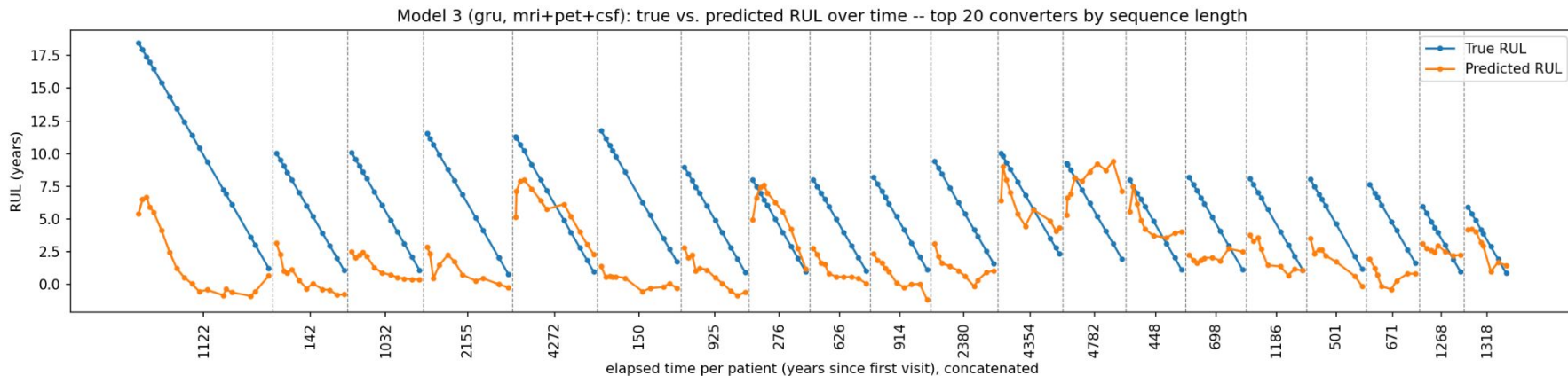
Time-to-conversion & Survival Analysis

How likely is a patient to convert to a worse diagnosis, and by when

Naive regression vs classification

Developed two families that model the same progression but answer to different questions:

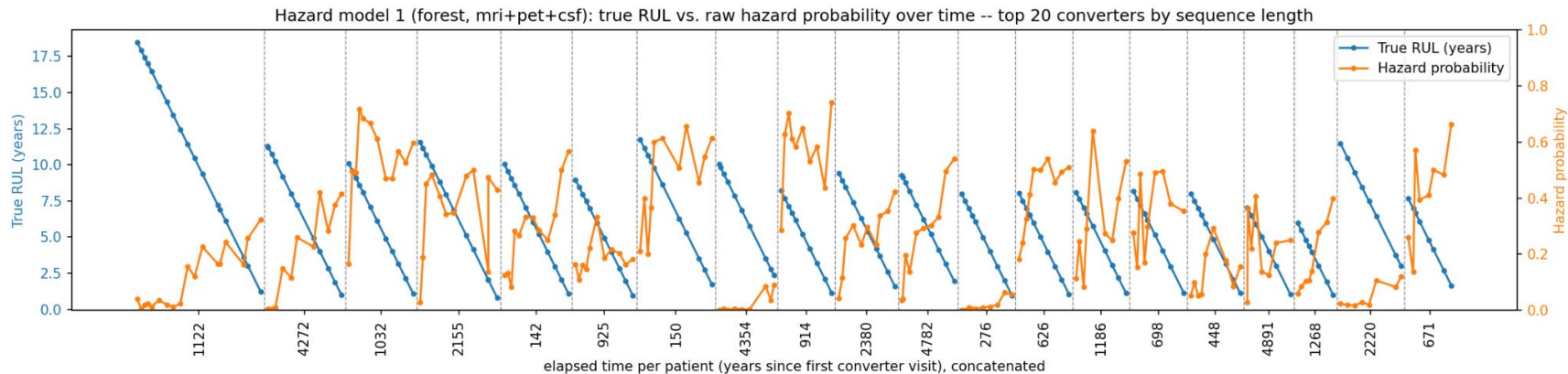
- **Regressors:** "How many years until this patient converts?". A direct numeric prediction of RUL_YEARS.
- **Classifiers:** "Is this visit the one immediately preceding conversion?". A binary EVENT_AT_VISIT label, true for exactly one visit per converter.



Naive regression vs classification

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Naive regression vs classification

	Regressors	Hazard classifiers
Label needs	exact known time-to-event	binary "conversion happens in the next visit"
Uses non-converters?	no	yes
Direct output	RUL in years	P(convert at next step history)
RUL still obtainable?	yes, directly	could be derived by walking the survival curve forward
Evaluation metric	MAE	AUC + calibration gap

Incremental feature engineering approach

Regressor	Classifier	Input shape
rul_model_1	hazard_model_1	single visit (level only)
rul_model_2	hazard_model_2	pair of visits (level + change + gap)
rul_model_3	hazard_model_3	full patient sequence (RNN)

RUL Regressors

Model	Estimator	MAE	Baseline MAE
1 — single visit	RF	1.654	1.735
2 — visit pair	RF	1.755	1.735
3 — sequence (GRU)	GRU	~1.70–1.84 (best: pet alone, 1.702)	1.735

Hazard classifiers

Model	Estimator	AUC	Calibration gap
1 — single visit	RF	0.795	0.086
2 — visit pair	RF	0.780	0.066
3 — sequence (GRU)	GRU	0.738–0.764 (best: pet alone, 0.764)	0.04–0.09

The reasonable choice

0.795

test AUC — Random Forest
Hazard model classifier on
mri+pet+csf

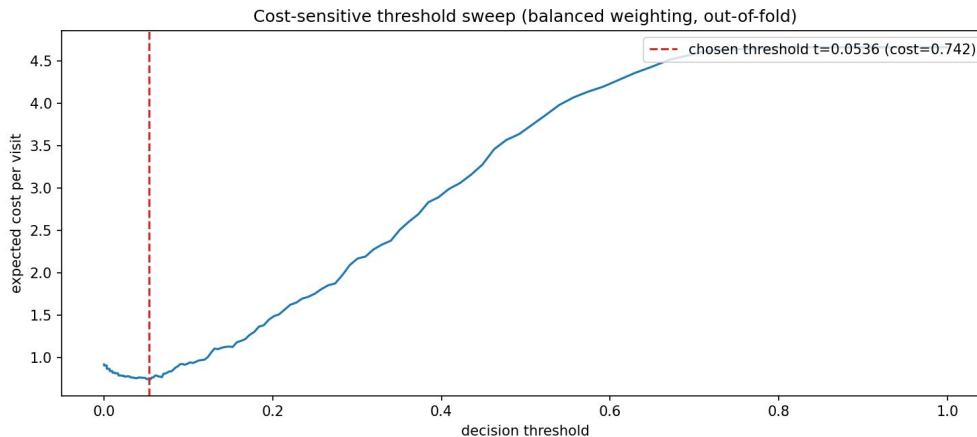
- **All three regressors land close to their baselines.** MAE improves over the mean-predictor baseline by roughly 5–10%, biomarkers add a modest signal. RUL prediction from these features alone is a hard problem on this cohort.
- **Combining all three modalities consistently beats any single modality**, in both regressors and hazard classifiers, MRI, PET, and CSF appear to carry at least partly complementary information.
- **Adding rate-of-change (model 2) or full sequence history (model 3) does not clearly beat the single-visit baseline (model 1).** The two likely reasons: PET/CSF are sparse and heavily filled for most visits, diluting the "change" signal; the added model complexity (MLP, RNN) needs more training rows.
- **Hazard classifiers substantially outperform a naive "predict the average" approach on discrimination** (AUC 0.74–0.79, well above the 0.5 chance level)
- **The hazard models are more data efficient.** By training on every visit of every patient (not just converters), the hazard models use 3–5x more rows than the matching regressors

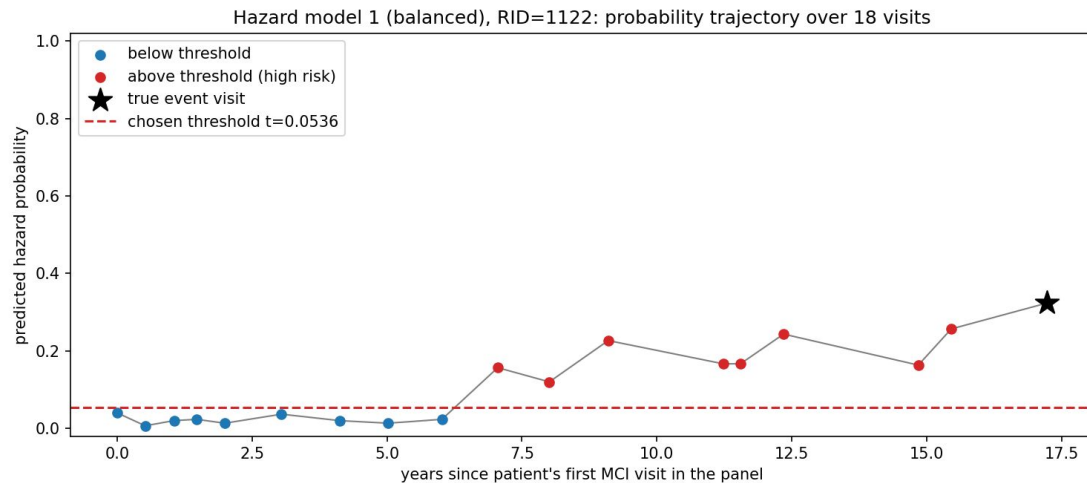
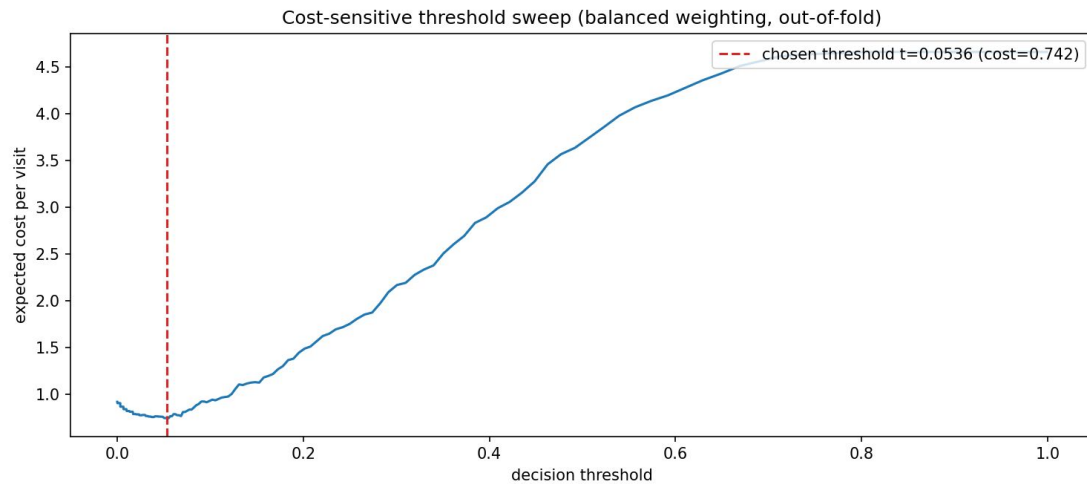
Ranking $\neq$ calibration

Taking into account the cost of missing conversions

Weighting	AUC	Calibration gap	Cost @ 0.5	Cost @ swept-best threshold
balanced	0.795	0.086	3.666	0.742
cost_sensitive	0.793	0.190	2.050	0.733

Cost weights coming from Decision Support





PART IV: POINT 3 & THE PIPELINE

Decision Support, then one real chain

Given a risk score, what should actually be done, and is that recommendation fair and explainable? This chapter focuses on two things: recommending a concrete interval for the patient's next visit, and giving a clear reason why that interval makes sense given the patient's specific risk factors.

Two tracks, one risk score

- **Track 1, the cost model.** ConversionCostModel, prices a monitoring interval against a routine check cost and a missed conversion cost stratified by diagnosis, using a 2.8 times literature anchored multiplier. Four interval policies were tested, each one reading more of a patient's history than the last, to see how much that extra history is actually worth.
- **Track 2, the fairness audit.** A discrimination index is computed across demographic attributes that are never a model input, they are used only to audit the output after the fact.

Interval policies compared

COST

```
cost = n_checks · c_routine +  
n_missed · c_miss(dx) · (Δt / ref_Δt)
```

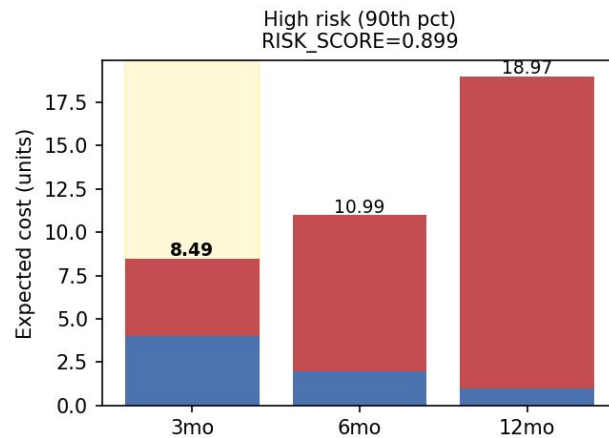
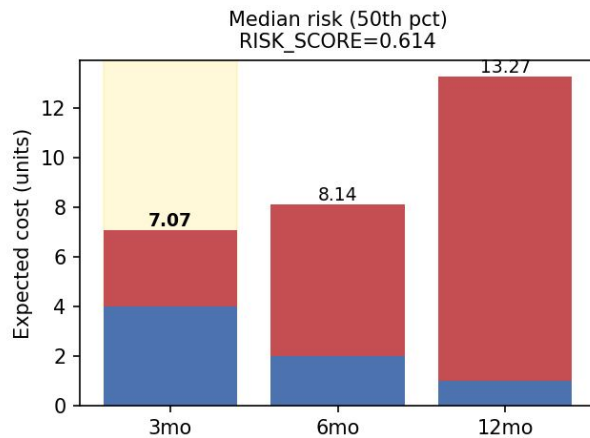
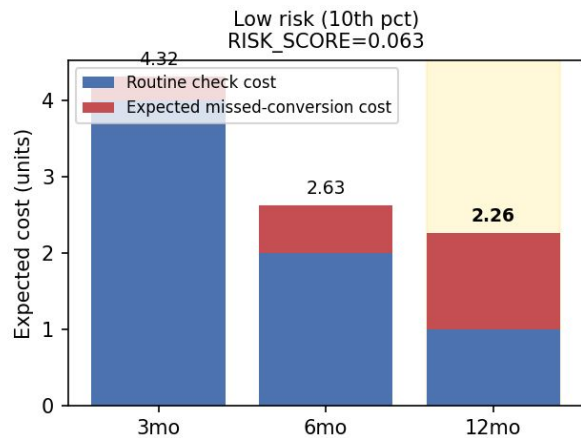
stratified by diagnosis, scales with interval length

CATCH RATE

```
catch_rate = caught_conversions /  
total_conversions
```

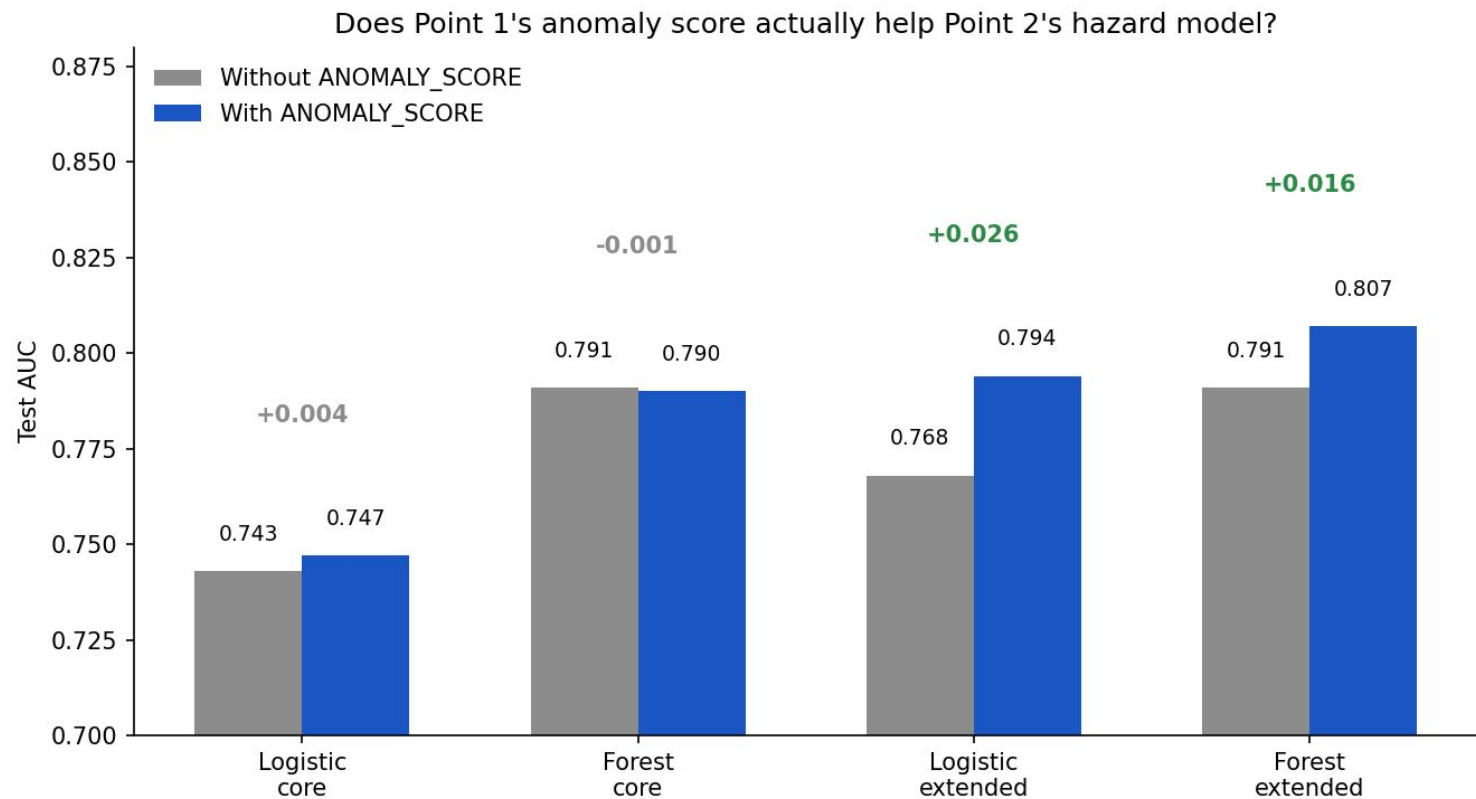
share of true conversions caught in time

Policy	Catch rate	Notable finding
Fixed (baseline)	95.2%	DIDI = 0 by construction, same interval for everyone
Snapshot / trajectory-adaptive	87–88%	Cost ~18,400–18,700 on test split
Forecast-adaptive	91.4%	Cost ~38,800 which is ~2× the others



Point 2 and 3 already work on their own

- Before Point 1 enters the picture at all, Point 2's hazard model, a soft fallback cascade across its core and extended feature tiers, already scores 69.9% of panel rows and drives Point 3's interval policy end to end on its own: test cost 12,050.7, DIDI 8.058, catch rate 90.4%.
- This sets the real bar for chaining. The case for wiring Point 1 in has to be that it measurably helps, not that the system fails without it. It does help, as the chart below shows.



Independent vs. chained

SAME MATCHED POPULATION, BOTH RUNS

independent = [extended, core] tiers only

chained = [extended+anomaly, extended, core+anomaly, core]

the only difference between the two runs is whether anomaly-including tiers are ever tried

Policy	Test cost	Test DIDI	Test catch rate
Independent (Point 2+3 alone)	12,050.7	8.058	90.4%
Chained (+ Point 1's anomaly score)	11,621.7	7.695	89.0%

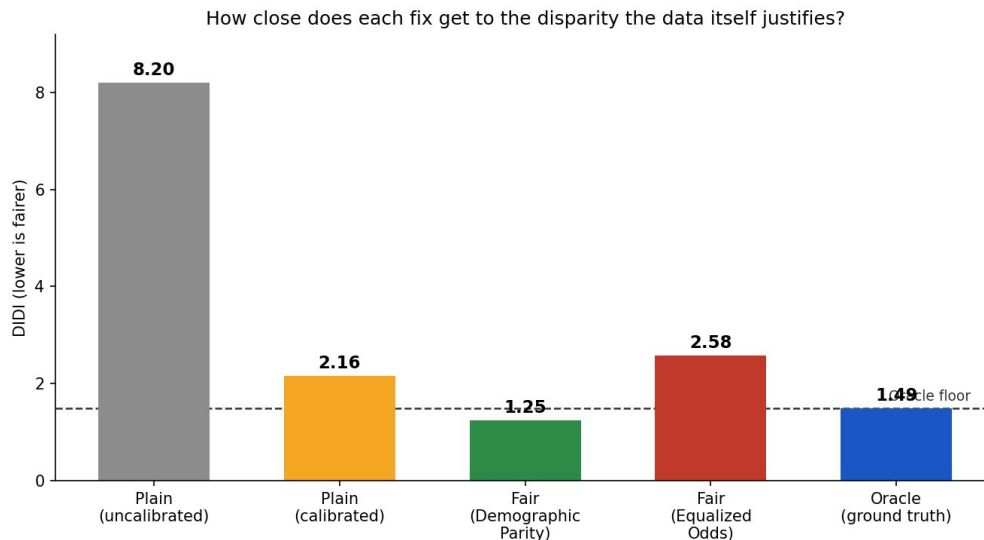
Same 69.9% population, same cost model as the independent baseline - a matched, single-run comparison.
Chaining is a modest, real win on cost and fairness (-3.6% cost, DIDI 8.058→7.695), at a cost of 1.4 points of catch rate.

The disparity created by the pipeline

DIDI: DISPARATE IMPACT DISCRIMINATION INDEX

$$\text{DIDI} = \sum_g \left| \text{avg}(\hat{y} \mid \text{group}=g) - \text{avg}(\hat{y}) \right|$$

lower is fairer, post-hoc = a cheap per-group score shift, ExpGrad = a genuine retrain under a fairness constraint



ExpGrad alone is the strongest fix (-71.3%). Layering the post-hoc shift on top gives most of the disparity back (2.208 → 6.155). ExpGrad alone is the mechanism actually deployed.

What this chapter establishes

- Point 3 turns a risk score into an actionable, audited, explainable decision.
- Chaining Point 1 in is a genuine, measured, modest win: 3.6% lower cost, better DIDI, 1.4 points of catch rate.
- A future improvement could be the use of Decision Focused Learning, to consider the minimization of a cost while training itself.

Thank you for your attention!